

ECE Navigation

Team 30

Basic Map Features

- Streets
- Zoom Level
- POIs and Features



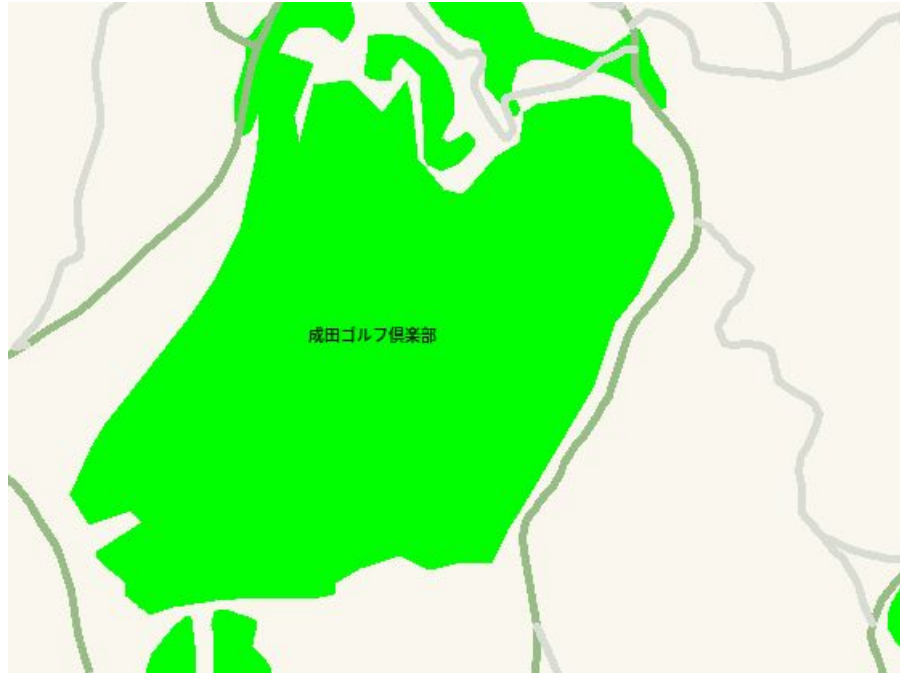
Extra Map Features

- Scale Bar
- Search Autocomplete



Extra Map Features

- Feature Layering
- Dark mode
- Language Integration



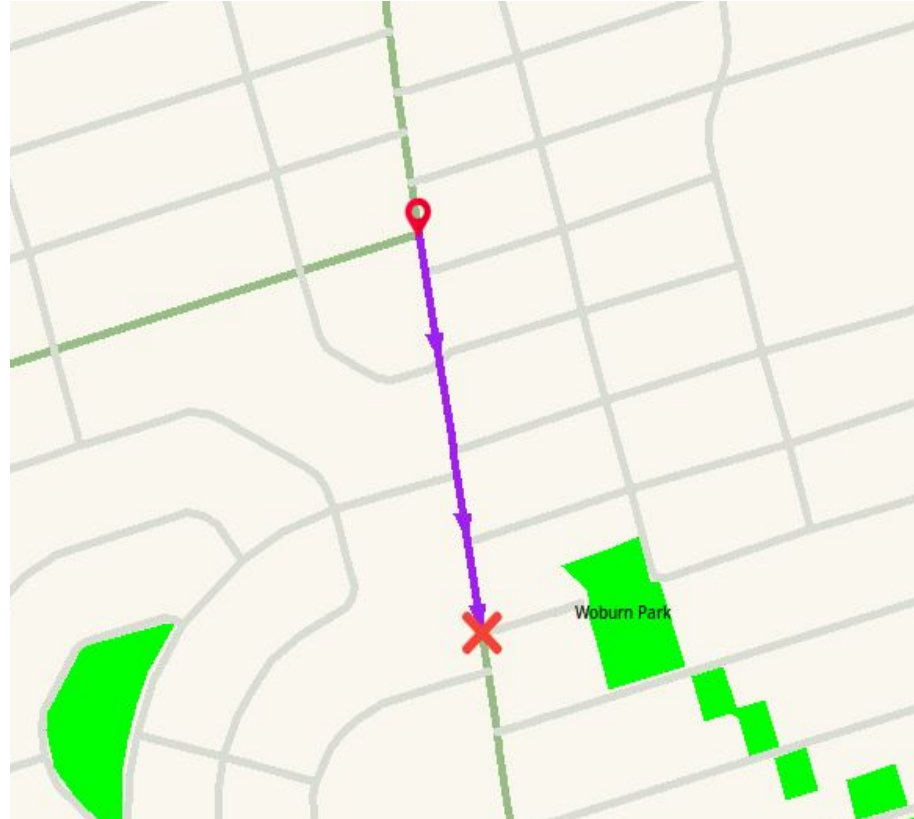
Path Finding

Dijkstra's Algorithm

Path Finding Algorithm

Searches Intersection to
Intersection

Keeps the Path with fastest time



A-Star Algorithm

Admissible Heuristic

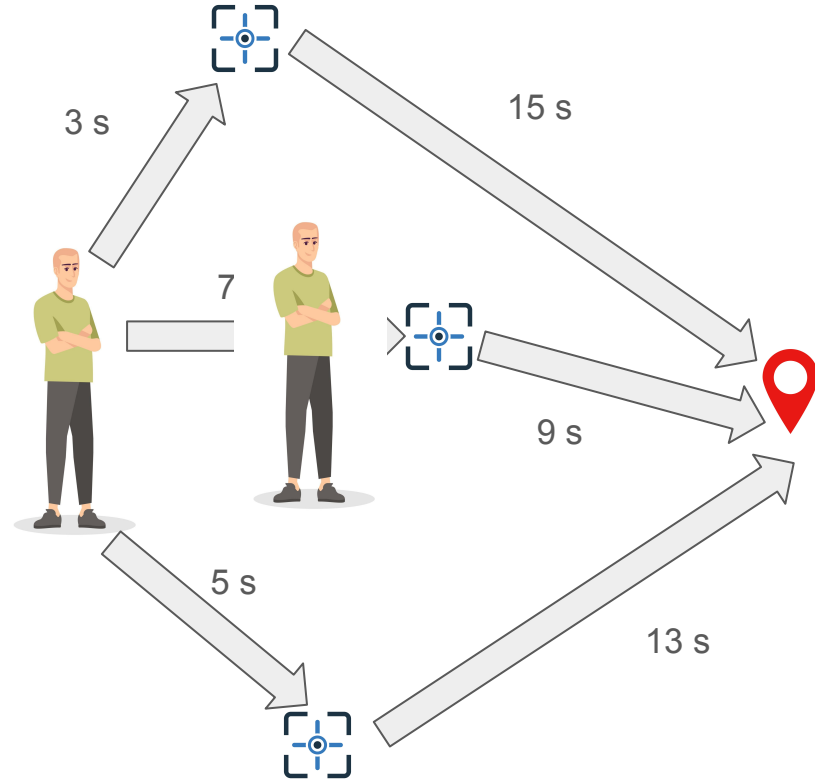
- Euclidean Distance

How does A-Star use the admissible heuristic?

- Time or Distance?
- How do we find the time?

Why's it any better than Dijkstra's?

- Less CPU time wasted
- Nodes are quickly “eliminated from contention”



Results

Test Summary: **PASS** (0 of 30 failed)

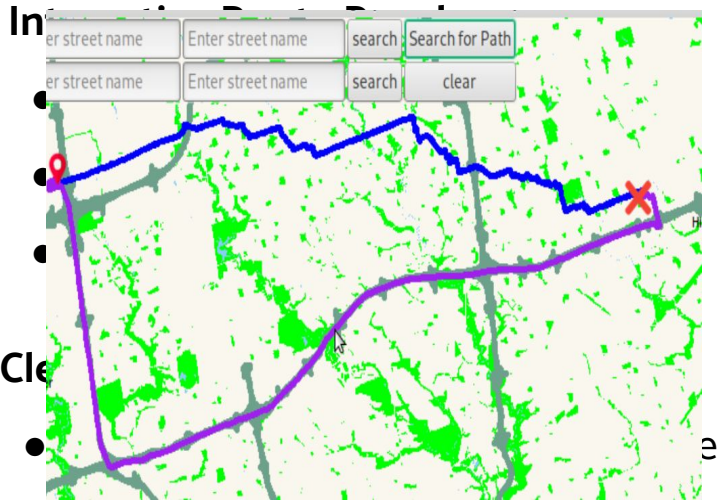
M3_Public_Func_Path_Cost_Interlaken_Switzerland	PASS	(0 of 1 failed)
M3_Public_Func_Inter_Inter_Simple_Legality_Interlaken_Switzerland	PASS	(0 of 1 failed)
M3_Public_Func_Inter_Inter_Simple_Optimality_Interlaken_Switzerland	PASS	(0 of 1 failed)
M3_Public_Func_Inter_Inter_Legality_Interlaken_Switzerland	PASS	(0 of 1 failed)
M3_Public_Func_Inter_Inter_Optimality_Interlaken_Switzerland	PASS	(0 of 1 failed)
M3_Public_Func_Path_Cost_Toronto_Canada	PASS	(0 of 1 failed)
M3_Public_Func_Inter_Inter_Simple_Legality_Toronto_Canada	PASS	(0 of 1 failed)
M3_Public_Func_Inter_Inter_Simple_Optimality_Toronto_Canada	PASS	(0 of 1 failed)
M3_Public_Func_Inter_Inter_Legality_Toronto_Canada	PASS	(0 of 1 failed)
M3_Public_Func_Inter_Inter_Optimality_Toronto_Canada	PASS	(0 of 1 failed)
M3_Public_Func_Path_Cost_London_England	PASS	(0 of 1 failed)

M3_Public_Func_Inter_Inter_Simple_Legality_London_England	PASS	(0 of 1 failed)
M3_Public_Func_Inter_Inter_Simple_Optimality_London_England	PASS	(0 of 1 failed)
M3_Public_Func_Inter_Inter_Legality_London_England	PASS	(0 of 1 failed)
M3_Public_Func_Inter_Inter_Optimality_London_England	PASS	(0 of 1 failed)
M3_Public_Valgrind	PASS	(0 of 7 failed)
M3_Public_Perf_Load_Map	PASS	(0 of 2 failed)
M3_Public_Perf_Inter_Inter_Medium_Toronto	PASS	(0 of 1 failed)
M3_Public_Perf_Inter_Inter_Hard_Toronto	PASS	(0 of 1 failed)
M3_Public_Perf_Inter_Inter_Very_Hard_Toronto	PASS	(0 of 1 failed)
M3_Public_Perf_Inter_Inter_Medium_London_England	PASS	(0 of 1 failed)
M3_Public_Perf_Inter_Inter_Hard_London_England	PASS	(0 of 1 failed)
M3_Public_Perf_Inter_Inter_Very_Hard_London_England	PASS	(0 of 1 failed)

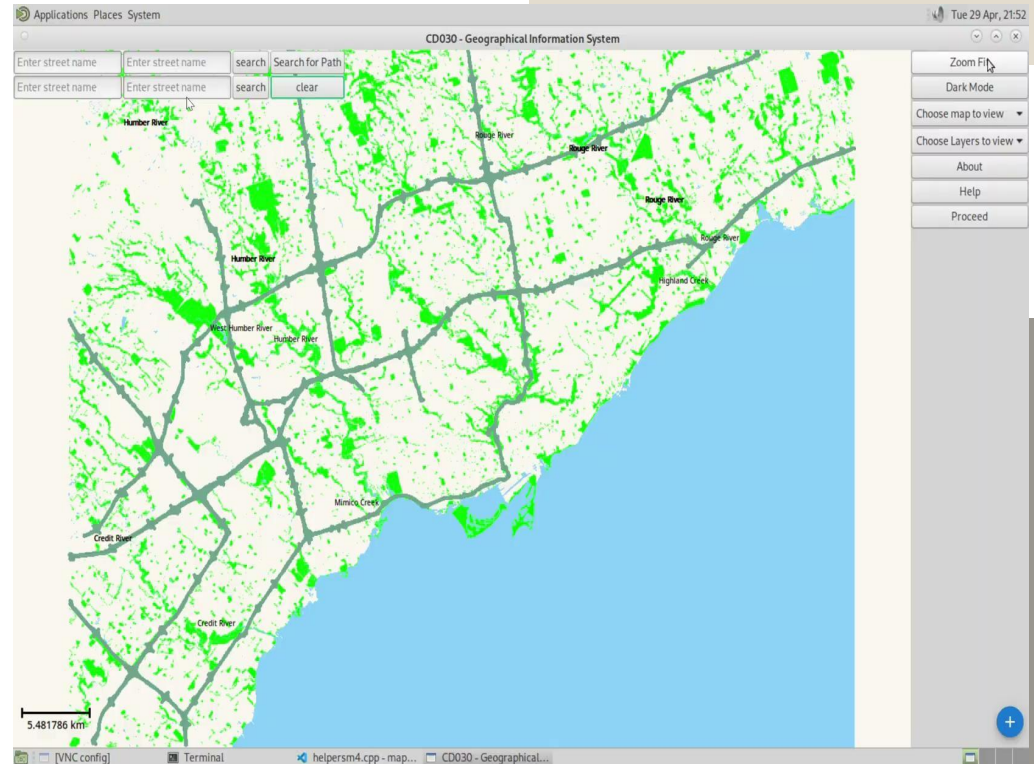
Route Visualization and UI Features

Separate route display

- Walking = blue
- Driving = purple



- Panel appears only with active route



Traveling Courier Problem

Traveling Courier Problem



1. Leave from a Depot
2. Pick up deliveries
3. Drop off deliveries
4. Go back to Depot

↖ Next candidate

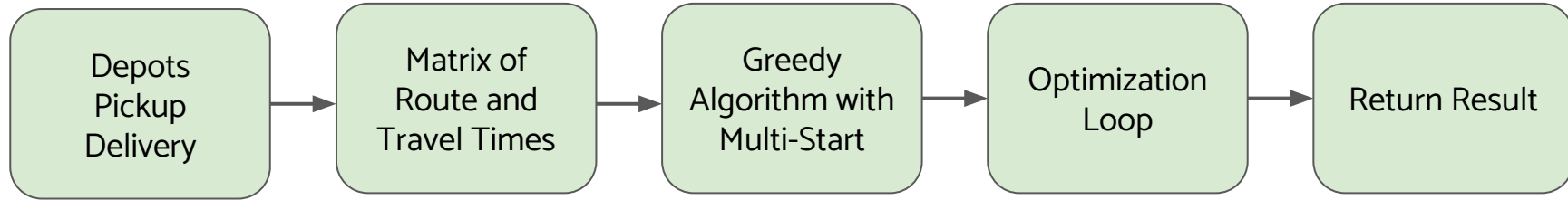
Legality Check for no rule violation

← Already visited?

No ➡ Not eligible to visit

Ensures “Pick-Up Before Drop-Off” in all cases

Problem “Solution”



Complex

Simulated Annealing Function

Genetic Algorithms



Simple

Change it slightly

Keep if it's better

Greedy Algorithm & Remake Route

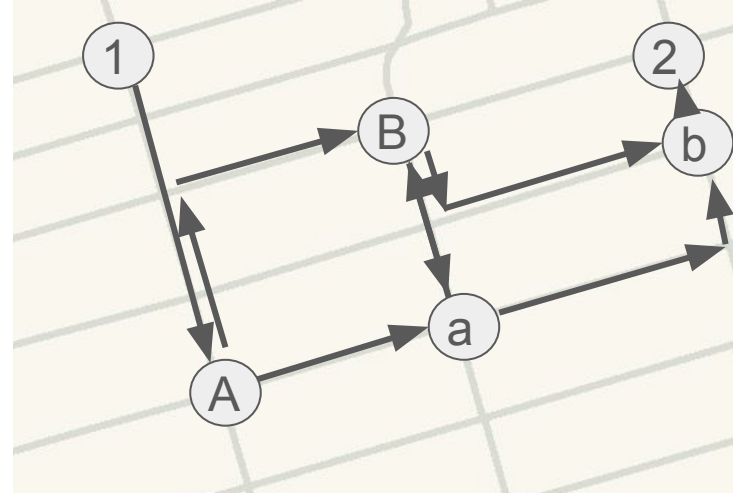
Used Greedy Algorithm to make Initial Route

A “Perturbing” function

- Slightly changing the route

Randomly take out a pickup-delivery pair

Randomly put it back



1	A	a	B	b	2
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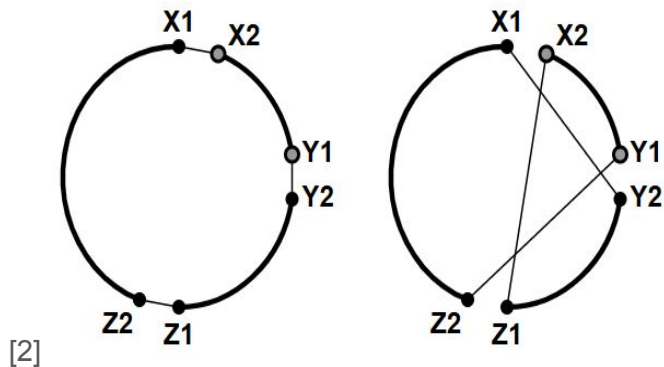
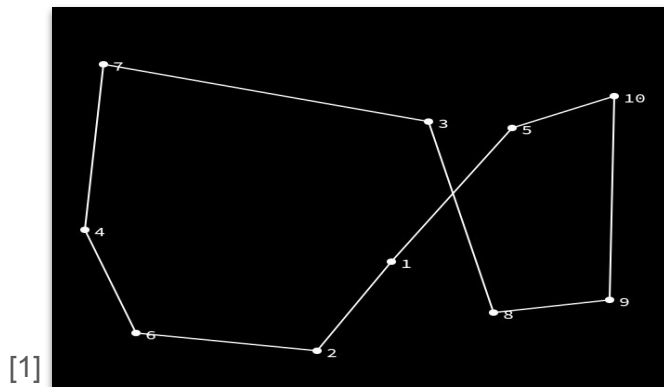
Simple Perturbations

2-Opt:

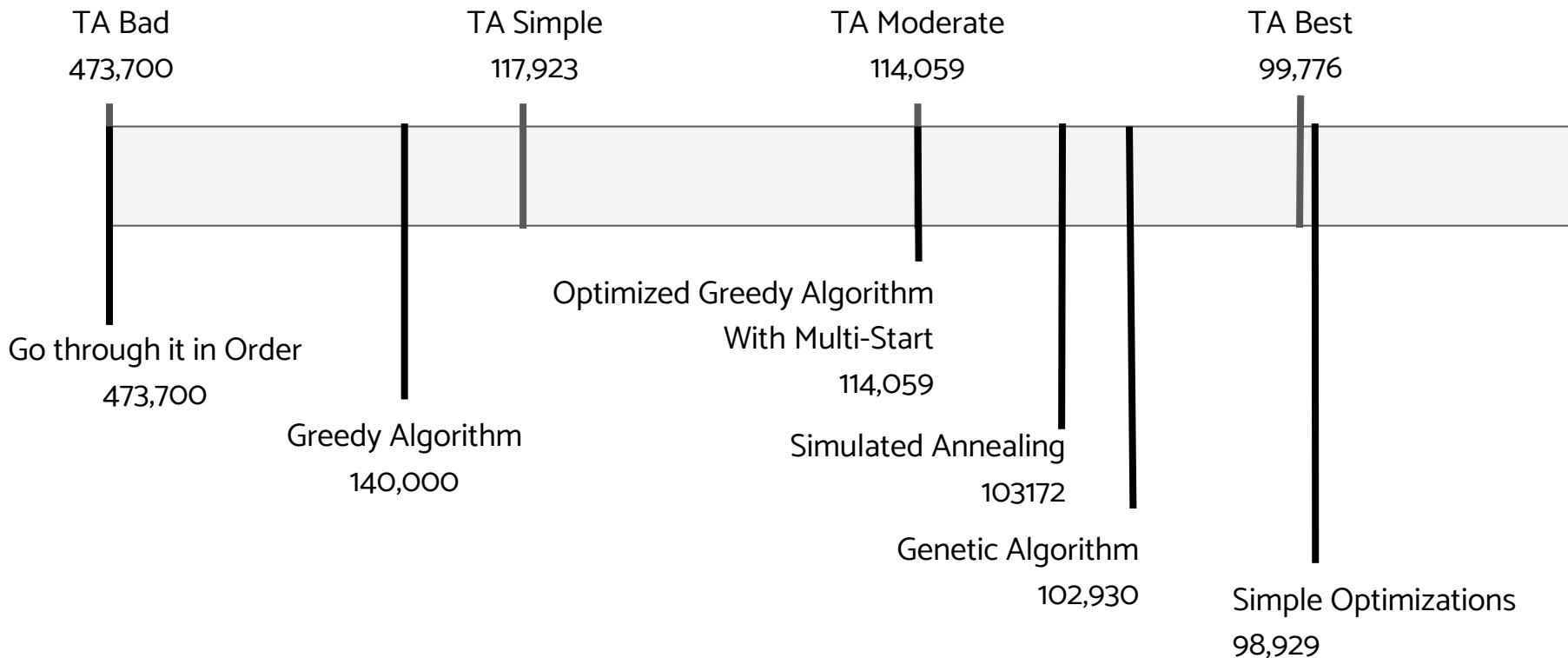
- “Delete” two edges
- Reconnect path in a new manner

Path Rotation/Segment Shifting:

- Choose a segment (2 nodes and the path between them)
- Rotate chosen sub-path to an earlier index in the complete path (collection of subpaths)



Quality of Results per Approach



Our Rank

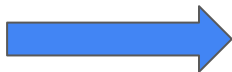
Rank	Group	Number of failed tests	M4 Quality of Result (QoR)	Percent Difference to Best	Percent Difference to Previous	Notes
1	an-133	0	90994.22	0.0 %	0.0 %	-
2	an-709	0	93520.675	2.78 %	2.78 %	-
3	an-622	0	93645.039	2.91 %	0.13 %	-
4	an-127	0	93925.122	3.22 %	0.3 %	-
5	an-469	0	94489.455	3.84 %	0.6 %	-
6	an-150	0	95245.084	4.67 %	0.8 %	-
7	an-667	0	95450.995	4.9 %	0.22 %	-
8	an-201	0	95856.709	5.34 %	0.43 %	-
9	an-942	0	97122.587	6.73 %	1.32 %	-
10	an-298	0	98111.329	7.82 %	1.02 %	-
11	an-200	0	98175.198	7.89 %	0.07 %	-
12	an-754	0	98209.679	7.93 %	0.04 %	-
13	an-484	0	98431.09	8.17 %	0.23 %	-
14	an-888	0	98929.703	8.72 %	0.51 %	-



Us!



Best TA Solution!



9	an-942	0	97122.587	6.73 %	1.32 %	-
10	an-298	0	98111.329	7.82 %	1.02 %	-
11	an-200	0	98175.198	7.89 %	0.07 %	-
12	an-754	0	98209.679	7.93 %	0.04 %	-
13	an-484	0	98431.09	8.17 %	0.23 %	-
14	an-888	0	98929.703	8.72 %	0.51 %	-
15	an-316	0	98986.699	8.78 %	0.06 %	-
16	an-313	0	99304.03	9.13 %	0.32 %	-
17	an-491	0	99529.774	9.38 %	0.23 %	-
18	an-876	0	99750.538	9.62 %	0.22 %	-
-	TA Best	0	99776.971	9.65 %	0.03 %	-
19	an-128	0	100218.808	10.14 %	0.44 %	-
20	an-683	0	100557.124	10.51 %	0.34 %	-
21	an-419	0	100963.523	10.96 %	0.4 %	-
22	an-641	0	101107.445	11.11 %	0.14 %	-

Plans for the Future

Filling the Gaps

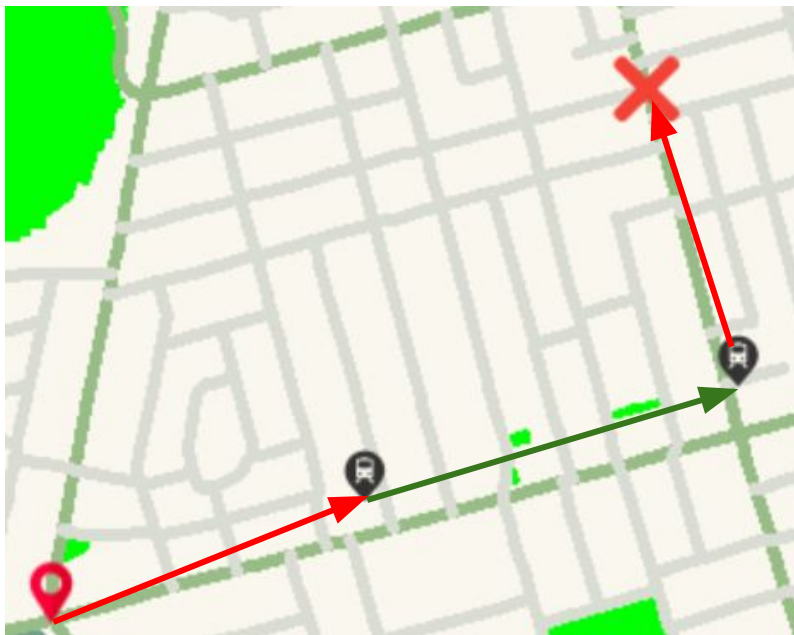
- The Gap Present between us and the Market

- Walking and Driving Directions Completed 
 - Subway/Walking Combination 
 - Road Closures 
- } We must work on these!

- The Gap within the Market

- Subway Delays rarely integrated into apps
- Emergency alerts as well
- Must also offer student-related functionalities: Learning Space Management & Sync Search

Combined Walking and Subway Routes

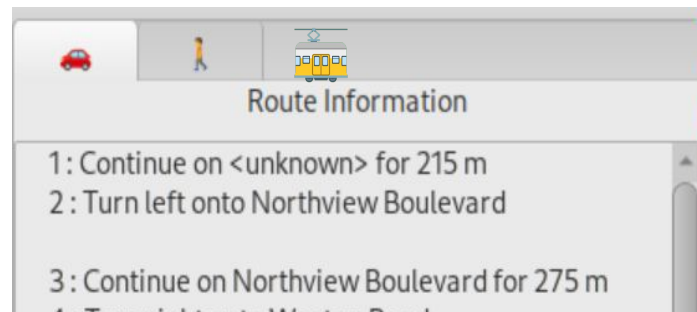


Find nearby stations

Connect by subway

Link by walking

➡ Easy navigation



Filling the Gaps

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Nobody's
done this
yet?
Okay,
we'll do it!

API Integration

Road Closures

Subway Delays

Emergency Alerts



Campus Safety Special Constable Service
Mar 21 • 2 min read



UPDATE: CSA 25-001: TPS News Release - MAN ARRESTED

Broadcast Time: 17:15 Date: March 21, 2025 Type:
Community Safety Alert (CSA) Homicide
Investigation, Queen's Park Crescent and College...
This is Toronto's 5th Homicide of 2025.

LSM + Sync Search and beyond!

Learning Space Management -
Images of buildings and their
lecture halls and rooms

Sync Search - Finding free study
spaces on Campus



Too many rooms!

Which one do I want?

BA 1210

Bahen Centre Information Tech



Apr 23, 2025 at 7:00 PM - 8:00 PM

Room Capacity: 85

Wheelchair Accessible:

[Room Homepage](#)

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Thank you for listening

Any questions?

References

- [1] Ole Kröger, “TSPSolver.jl: 2-opt,” *Opensourc.es*, 2019. [Online.] Available: <https://opensourc.es/blog/tspsolver.jl-2-opt/>. Accessed: Apr. 25, 2025.
- [2] “Shifting a segment,” *Blogspot.com*, 2017. [Online.] Available: <https://tsp-basics.blogspot.com/2017/03/shifting-segment.html>. Accessed: Apr. 25, 2025.
- [3] “Toronto traffic: Ongoing construction closures across the GTA,” CityNews Toronto, [Online.] Available: <https://toronto.citynews.ca/traffic/>. Accessed: Apr. 29, 2025.
- [4] “Service alerts,” TTC.ca, [Online.] Available: <https://www.ttc.ca/service-alerts>. Accessed: Apr. 29, 2025.
- [5] “University of Toronto,” Campus Safety, [Online.] Available: <https://www.campussafety.utoronto.ca/>. Accessed: Apr. 29, 2025.
- [6] “Learning Space Management, University of Toronto,” *lsm.utoronto.ca*, 2024. [Online.] Available: <https://lsm.utoronto.ca/index.html>. Accessed: Apr. 23, 2025.
- [7] “ECE297 Lecture slide 23B, University of Toronto,” https://q.utoronto.ca/courses/376139/pages/lecture-20-m4-complexity-and-greedy-heuristics?module_item_id=6632323. Accessed: Apr. 29, 2025.